

What motivates giving in Dictator Games?

Estimating the role of social norms and preferences

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Abstract

Altruistic cooperation is well documented in laboratory and field settings, yet the motivations that sustain it remain debated. Two leading explanations emphasize outcome-based social preferences and social norms, with the former focusing on distributional outcomes and the latter on compliance with socially appropriate action. We systematically study the relative importance of these competing explanations by conducting structural estimation on a dataset constructed by combining individual-level data from Dictator Game experiments. Additionally, we study the role of social norm strength—the extent to which social norms are clearly defined and widely shared—in translating what should be done into action. We find that generosity in Dictator Games is jointly determined by outcome-based social preferences and social norms, and that the influence of social norms on behaviour is strongly associated with their strength.

JEL codes: D90, D91

Keywords: Altruism, social preferences, social norms, meta-analysis, structural estimation

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1 Introduction

Over the past 40 years, a large and compelling experimental literature has shown that people are willing to benefit others even at a net cost to themselves (Fehr and Gächter [2000], Engel [2011], Henrich et al. [2010]). While such altruistic cooperation is well established, and multiple proximate motivations have been proposed (Andreoni [1990], Fehr and Schmidt [1999], Bolton and Ockenfels [2000]), their relative contributions to driving cooperation remain unclear. We address this issue by structurally estimating and comparing two key classes of explanations—outcome-based social preferences and social norm-based motivations—using a dataset combining individual-level data from Dictator Game experiments. Drawing on recent theoretical developments, we also test empirically the role of social norm strength (Gelfand et al. [2011], Szekely et al. [2021, 2025]) which aims to capture how strongly shared expectations about appropriate behaviour shape action.

Some of the strongest evidence for altruistic giving comes from Dictator Games, in which a “dictator” receives an endowment and decides how much, if any, of that to give to a passive receiver. Meta-analytic evidence (Engel [2011], Camerer [2011]) shows that a majority of people behave altruistically, many opting for a fair-split option, and only a minority behaving purely selfishly. A small proportion even give away more than they keep (see Engel [2011]). Varying the framing of the situation by making generosity less appealing, as in the bully variant of the Dictator Game (Krupka and Weber [2013]), does change the absolute levels of giving but does not eliminate the basic result: a substantial proportion of people are willing to give money to others at a cost to themselves.

Early attempts to explain these behavioural patterns focused on the role of preferences and expanded agents’ utility functions with *social preferences* that integrate the payoffs of others in addition to the standard self-interest component (Charness and Rabin [2002]). Most of these models assume that agents are concerned about the final distribution of outcomes; among the most influential theories implementing this approach are inequality aversion models (Fehr and Schmidt [1999], Bolton and Ockenfels [2000]). Under this outcome-based framework, people dislike inequitable distributions of earnings, whether this is advantageous or disadvantageous inequality, and prefer outcomes that lead to equal payoffs between parties.

More recently, precise accounts have also emerged that focus on the role of *social norms* in motivating giving behaviour (Bicchieri [2006], Krupka and Weber [2013]) and conceptualise social norms as informal rules governing behaviour in social interactions that are sustained by shared expectations of compliance and potentially sanctioning. Formally, these models integrate a desire to take the socially appropriate action into agents’ utility functions. Most parsimoniously, Krupka and Weber [2013] propose a model that captures both self-interest and a desire to behave in a socially appropriate manner. In the context of Dictator Games, subjects are torn between pursuing their own self-interest by giving nothing and complying with the social norm prescribed in the game, often the 50-50 split. These models focus on the *content* of social norms, the behaviour that is prescribed by the social norm and ought to be taken.

While extensive evidence indicates that social norms can shape cooperation, people do not follow them blindly either. Indeed, recent work argues that a key factor moderating whether normative pressure turns into behaviour is *social norm strength* (Szekely et al. [2025]). Weak social norms are inaccurate (e.g. when expectations do not match others’ behaviour), non-specific, and show little consistency within a group, while strong ones display the opposite properties (Szekely et al. [2021]).

The key implication is that strong social norms are closely followed, while weaker ones act more like gentle suggestions, easily prone to being ignored—especially when contradicting self interest. Although fragments of existing evidence point to an important role of social norm strength (e.g. Szekely et al. [2021, 2025], Gelfand et al. [2011], Lane et al. [2024]), systematic evidence is missing. To address this, here we provide the first comprehensive test of how social norm strength affects cooperative behaviour in Dictator Games.

Social preferences, social norms, and self interest have all been shown to influence giving behaviour. Self-interested motivations are apparent in the actions of the subjects who keep all of their endowment (e.g. Engel [2011]). Social preferences are also clear as some individuals give to others even when they observe that others around them do not (e.g. Bicchieri and Xiao [2009]). These preferences can also be reliably measured (Murphy et al. [2011], Murphy and Ackermann [2014], Nunnari and Pozzi [2026]) and they can be empirically distinguished from social norms (e.g. Szekely et al. [2023], Tverskoi et al. [2023]). However, while previous studies have estimated outcome-based and norms-based models separately, a comprehensive comparison of their relative contributions to explaining generosity is still missing. We contribute to this literature by constructing a dataset of Dictator Games from 12 studies (14 experiments, 1558 subjects) in which social norms are measured using the established incentivised method of Krupka and Weber [2013], and by structurally estimating and comparing outcome-based and norm-based models of altruistic giving while examining how social norm strength moderates the relationship between norms and behaviour. Our analytical approach follows that of Carpenter and Robbett (2022) and Chang et al. (2019), allowing us to estimate representative-agent preferences from choice data.

Our results show that generosity in Dictator Games is jointly shaped by social preferences and social norms, with social norm strength playing a crucial moderating role. When estimated in isolation, the representative agent displays altruistic preferences when in an advantageous position but behindness aversion when behind. However, once norm content enters the utility function, the role of social preferences nearly vanishes, indicating that much of what appears to be outcome-based altruism is in fact norm compliance. Social norm adherence is large and robust across specifications, with individuals willing to sacrifice substantial monetary payoffs to take an action deemed more socially appropriate. Finally, norm strength amplifies this relationship: behaviour is more tightly bound to norms when those norms are widely shared and clearly defined.

Our work closely relates to a group of papers structurally estimating social preferences. Bellemare et al. [2008] provide first structural estimates of inequity aversion in ultimatum games. Bruhin et al. [2019] use finite mixture models to classify subjects into types. Consistent with our own results, they find that most subjects are either strong or moderate altruists and that some display behindness aversion, (i.e. dislike others being better off, but indifferent to those worse off). Most recently, Nunnari and Pozzi [2026] provide meta-analytic evidence on distributional preferences, reporting an average advantageous-inequality aversion of $\hat{\beta} = 0.326$ across studies that do not separately elicit norm content. Our norm-free estimates are somewhat above this benchmark, while our norm-adjusted estimates fall well below it, suggesting that a portion of the distributional preferences documented in the literature reflects norm compliance rather than intrinsic other-regarding preferences. Other studies, including Krupka and Weber [2013] and Chang et al. [2019], provide structural evidence on the importance of norm adherence in Dictator Games. Carpenter and Robbett [2022] incorporate norm compliance directly into agents' utility functions in a single study and show that the representative agent's weight on the payoff of worse-off recipients declines from

$\hat{\beta} = 0.336$ to 0.263 once norms are controlled for. We document a qualitatively identical collapse at the meta-analytic level, with the preference weight falling from $\rho = 0.50$ to $\rho = 0.06$ once norm content enters the utility function. We build on this emerging literature by providing meta-analytic evidence of the role played by social preferences and social norms across multiple studies and by incorporating social norm strength into the utility of the representative agent.

While both outcome-based and norm-based frameworks can explain altruistic giving, they differ in their assumptions. Outcome-based models assume that individuals care about the distribution of payoffs; norm-based models emphasise conformity to socially prescribed behaviour. These differences have deep implications for behaviour change: outcome-based accounts suggest that long-term processes shaping preferences are essential for promoting cooperation (Bowles and Polania-Reyes [2012], Bowles [2016]), whereas norm-based accounts suggest that changing expectations about others' behaviour and social approval can have immediate effects (Bursztyn et al. [2020], Bicchieri and Xiao [2009], Szekely et al. [2021]), particularly when norms are strong (Szekely et al. [2025]). By jointly estimating these motivations and incorporating variation in norm strength, our framework provides a more complete account of how social norms shape prosocial behaviour.

The rest of this work is organized as follows. Section 2 outlines the models we structurally estimate. Section 3 describes methods and empirical strategy. Section 4 presents and discusses the results. Section 5 discusses the implications. Lastly, Section 6 concludes.

2 Conceptual framework

In this section, we specify the classes of utility models that we estimate to account for the choices made by dictators (D). These all draw on, or directly implement, established frameworks. The common setup to all models assumes that player D chooses an action from the action space $A = \{a^1, a^2, \dots, a^E\}$. Any given action a^k represents an allocation between herself and a receiver R , denoted as $\Pi = \{\pi_D^k, \pi_R^k\}$.

2.1 Selfish preferences

We start with the simplest model specification that standard economic theory has traditionally assumed and consider that D is uniquely motivated by their own monetary payoffs:

$$U_D(\pi_D^k) = \delta \cdot \pi_D^k \quad (\text{Eq. 1 - Selfish Model})$$

where π_D^k is the payoff associated with action a^k , and δ represents the weight that D attributes to the monetary payoff. With $\delta > 0$. This model specification uniquely predicts that subjects should give nothing and represents our benchmark for later comparisons.

2.2 Social preferences

While monetary payoffs represent an important factor affecting dictators' decisions, longstanding evidence in behavioural economics has shown that people care about other's payoffs, that is, they have social preferences (Fehr and Schmidt [1999], Andreoni [1990], Bolton and Ockenfels [2000], Charness and Rabin [2002]). In the context of Dictator Games, the underlying assumption is that

D cares about the distribution of payoffs between her and R . One of the most influential, and general, specification of social preferences in typical two-player games comes from Charness and Rabin [2002]:

$$U_D(\pi_D, \pi_R) = (\rho \cdot \mathbb{1}_{\pi_D > \pi_R} + \sigma \cdot \mathbb{1}_{\pi_D < \pi_R}) \cdot \pi_R + (1 - \rho \cdot \mathbb{1}_{\pi_D > \pi_R} - \sigma \cdot \mathbb{1}_{\pi_D < \pi_R}) \cdot \pi_D \quad (\text{Eq. 2 - Outcome Based})$$

This model specification captures a range of distributional preferences (such as inequity aversion models as in Fehr and Schmidt [1999]) that rely on final monetary outcomes to determine utility.¹ The parameters ρ and σ represent the weight that the D gives to the R 's payoff, when D earns, respectively, more or less. For example, if people are generally altruistic, hence willing to give up part of their payoff to increase the payoff of someone else, the model assumes $\rho, \sigma > 0$. However, if people are characterized by “behindness aversion”, this translates into $\sigma < 0$. Similarly, if people are “aheadness averse”, their $\rho > 0$. As a consequence, if people show concern for differences in distributions (inequity aversion), the model assumes that $\sigma < 0 < \rho$.

2.3 Social norms

More recent experimental evidence has demonstrated that individuals care not only about the absolute outcomes that they receive from interactions but also whether the actions that led to those outcomes are consistent with social expectations of appropriate behavior, or *social norms* (Bicchieri [2006], Krupka and Weber [2013], Szekely et al. [2021]). As models of social preferences are not able to account for these behavioral regularities, we draw on models that directly integrate social norms in agents' utility function. While several authors have suggested different model specifications (e.g. Bicchieri [2006]), we consider the specification introduced by Krupka and Weber [2013]. This specification posits that social norms indicate the extent to which action are socially appropriate, and, because of the incentivized elicitation method used to identify them empirically, they provide a sound foundation for structural estimation. In the case of standard Dictator Games, D 's utility that considers social norms is defined as:

$$U_D(\pi_D, N_k) = \delta \cdot \pi_D + \gamma \cdot N_k \quad (\text{Eq. 3 - Social Norm Content})$$

where N_k is a value in the interval $[-1, 1]$ representing the social appropriateness of action a^k . The action a^k with the highest appropriateness score $N(a^k)$ represents the *content* of the norm, that is the socially prescribed action. Parameter γ reflects the degree to which person cares about complying with social norms (individual sensitivity to social norms). In this model, the degree to which a person cares about adhering to any social norm is fixed. Intuitively, if an individual is characterized as a “norm follower” then she will typically be a “norm follower” in any situation.

We are also interested in comparing a mix of the two previous models, social preferences based (Eq. 2) and social norm based (Eq. 3) to directly compare the explanatory power of social preferences

¹While the extended formulation of the Charness and Rabin model also considers reciprocal motives, we put these to one side as they cannot play a role in standard one-shot Dictator Games. For the same reason, we do not consider social-welfare models (Bolton and Ockenfels [2000]) as these social preferences do not play a role in the context of the standard version of the game.

and social norms. The combined model that we estimate includes both outcome-based and social norms into D 's utility:

$$\begin{aligned}
U_D(\pi_D, \pi_R, N_k) = & (\rho \cdot \mathbb{1}_{\pi_D > \pi_R} + \sigma \cdot \mathbb{1}_{\pi_D < \pi_R}) \cdot \pi_R \\
& + (1 - \rho \cdot \mathbb{1}_{\pi_D > \pi_R} - \sigma \cdot \mathbb{1}_{\pi_D < \pi_R}) \cdot \pi_D \\
& + \gamma \cdot N_k
\end{aligned} \tag{Eq. 4 - Combined Model}$$

2.4 Social norm strength

Beyond recognising social norms as a key factor in explaining why individuals act prosocially, recent research has emphasized that it is not only the content of social norms that matter, but also how strong that norm is (Gelfand et al. [2011], Szekely et al. [2021, 2025]). Strong norms—those that are widely shared, clearly defined, and socially enforced—should be likelier to predict consistent and sustained prosocial behaviour, while weak norms—marked by ambiguity and uncertainty—should be associated with lower compliance and greater behavioural variability. In brief, norm strength may influence the relationship between social expectations and prosocial decisions in Dictator Games.

Social norm strength, in its most detailed form, has been conceptualised and operationalised along three dimensions: accuracy (whether beliefs match observed behaviour), consistency (the extent to which beliefs are shared within a group), and specificity (the range of appropriate actions) (Szekely et al. [2021]). Even from a broader perspective (Szekely et al. [2025]), there is general agreement that stronger social norms should be shared closely within a group and be restrictive in permitted actions (e.g. Gelfand et al. [2011], Szekely et al. [2021], Przepiorka et al. [2022], Lane et al. [2024], Dimant [2023]). To capture this complex idea in a parsimonious way, we focus on the role of variability in social expectations about appropriate behaviour. Greater variability indicates a weaker social norm while less variability indicates a stronger social norm. Specifically, we compute the standard deviation of social appropriateness ratings elicited using the Krupka and Weber [2013] method for each of the elicited actions in the Dictator Game, and rescale it so that higher values correspond to stronger (less dispersed) norms.

More specifically, for an action k , given a group of n agents, let

$$\sigma_k = \sqrt{\frac{1}{n} \sum_i (N_{ik} - \bar{N}_k)^2}$$

be the standard deviation of appropriateness ratings for action k . We then define social norm strength as

$$S_k = 1 - \frac{2 \sigma_k}{\max_j \sigma_j}$$

where the maximum is taken over all actions and treatments in the sample. S_k lies in $[-1, 1]$, with higher values indicating a more strongly shared (less dispersed) norm.

We then estimate a model that includes social norm strength:

$$\begin{aligned}
U_D(\pi_D, \pi_R, N_k, S_k) = & (\rho \cdot \mathbb{1}_{\pi_D > \pi_R} + \sigma \cdot \mathbb{1}_{\pi_D < \pi_R}) \cdot \pi_R + \\
& (1 - \rho \cdot \mathbb{1}_{\pi_D > \pi_R} - \sigma \cdot \mathbb{1}_{\pi_D < \pi_R}) \cdot \pi_D + \\
& \gamma \cdot N_k + \eta \cdot S_k
\end{aligned} \tag{Eq. 5 - Social Norm Strength}$$

While Eq. (5) allows social norm strength to enter utility as an additive component, our claim is stronger and proposes that social norm strength affects whether any given level of appropriateness translates into action. To account for this, we also allow norm content and norm strength to interact in utility; Eq. (6) captures the intuition that strong norms operate as focal behavioral constraints, whereas weak norms function more like non-binding suggestions.

$$\begin{aligned}
U_D(\pi_D, \pi_R, N_k, S_k) = & (\rho \cdot \mathbb{1}_{\pi_D > \pi_R} + \sigma \cdot \mathbb{1}_{\pi_D < \pi_R}) \cdot \pi_R + \\
& (1 - \rho \cdot \mathbb{1}_{\pi_D > \pi_R} - \sigma \cdot \mathbb{1}_{\pi_D < \pi_R}) \cdot \pi_D + \\
& \gamma \cdot N_k + \eta \cdot S_k + \nu \cdot N_k \cdot S_k
\end{aligned} \tag{Eq. 6 - Full}$$

3 Methods and empirical strategy

3.1 Search strategy and dataset

We conducted a keyword search across several search platforms, such as Google Scholar, EconRepec, WebOfScience, Ideas. We also sent post public announcement to the Economic Science Association (ESA) group asking for references and data. In our systematic search, we used all combinations of the keyword “dictator game” and the keywords “norm elicitation”, “social norms” and “norm expectation”. We then conducted a first screening of all experiments to ensure that they satisfied two main criteria. First, participants played one or more incentivized Dictator Games with human counterparts (i.e., the counterpart is not a computer or an hypothetical person). Second, social norms were elicited using the method in Krupka and Weber [2013]. While social norms can be elicited through multiple methods, the Krupka & Weber approach is most directly suited to utility estimation. This search gave an initial result of 33 papers.

We then classified papers with respect to their design and treatments. Our aim was to select an homogeneous group of games that are comparable in key study features. In particular, we collected and noted information about the *experimental design*, e.g. choice method (direct or strategy method), presence of punishment, rewards, laboratory or online study; *beliefs and expectations variables*, that is information regarding the elicitation of social norms (e.g., within subjects or between subjects), and the use of monetary incentives in the social norm elicitation.

After gathering all relevant information from the studies, we applied a second filter to select Dictator Games that matched the criteria below:

- Full endowment allocation to the dictator: We excluded studies where the initial endowment was split equally between the dictator and the receiver (e.g., the “bully” variant in Krupka and Weber [2013] or all Take-or-Give games).
- No multipliers: We retained only games where the amount transferred by the dictator to the receiver was not subject to any multiplication or transformation.

- Neutral framing: We included only studies in which decisions were made in a context-free, neutrally framed environment, avoiding moral or emotionally charged language.
- Real monetary incentives: We only considered studies with actual monetary payment for both choices in the Dictator Game and norm elicitation task.

This reduced the number of eligible studies to 25. Among these, we were able to collect individual-level data from the authors of 12 papers, 14 treatments, for a total of 1558 participants. All included experiments were run in laboratory settings. The full list of papers included and their main characteristics are reported in Table 1. 9 of the included treatments measures social norms with a distinct sample of participants than those who made decisions in the DG (*Between-Subjects*, see column in Table 1), while the remaining 5 measure norms with the same participants (*Within-Subjects*, Table 1). Other studies’ heterogeneity in their design aspects are discussed in Appendix A2 to show comparability among studies.

Table 1: Summary of Major Variables per Game Type

Paper	Treatment Label	Between- or Within-Subjects Norm elicitation	N. Subjects
List [2007] ²	Baseline	B	24
Lazear et al. [2012]	No Sorting	B	135
Krupka and Weber [2013]	Baseline	B	52
Kimbrough and Vostroknutov [2016]	DG	B	66
Della Valle and Ploner [2017]	Baseline	W	29
Gächter et al. [2017]	NoPeer Give	B	71
Gächter et al. [2017]	NoPeer Take	B	71
Thomsson and Vostroknutov [2017]	Baseline	W	53
Chang et al. [2019]	Neutrally-framed	B	56
Fromell et al. [2019]	Experiment 2 (Control)	B	126
Bašić and Verrina [2021]	Private - DG	W	127
Erkut [2022]	GENExperiment-Money	B	38
Eckel et al. [2023]	Class	W	357
Eckel et al. [2023]	Freshmen	W	353
Total			1558

3.2 Empirical strategy

We structurally estimate individuals’ preference parameters using a standard random utility model (McFadden [1974]). In doing so, we estimate individual utility functions to choice data using conditional logistic regressions where the dependent variable is the selected action. In this model, variation reflects the variability across characteristics of the possible actions.³ Given an action a^k ,

³As noted in Chang et al. [2019], conditional logistic models are similar to multinomial logistic models. However, conditional logistic models emphasize the characteristics of the alternatives, while multinomial logistic models depend on the characteristics of the individual making the choice. See Hoffman and Duncan [1988] for a comparison of these models.

each player i employs a logistic choice rule such that the probability P_i of choosing the action a^k is defined as:

$$P_i(a_i = a^k) = \frac{\exp(u_i(a^k))}{\sum_k \exp(u_i(a^k))}$$

The specification of u varies according to the model considered. In all of our analyses, we will estimate the utility parameters assuming that there is a representative agent for the whole population. We provide estimates from models for each of the papers selected in our database.⁴ To compare model fits, we use the Aikake Information Criteria (AIC). In particular, smaller values of this indicator are reflective of a better fit of the model to the data.

4 Results

4.1 Choices and social norms

The pooled average of amounts sent across all the dictators games considered is around 25% of the initial endowment. A good fraction of subjects, around 30%, send nothing, while over 25% send some amount near the fair split. A small fraction of subjects send more than the half of the initial endowment. While all of these results are similar to those typically observed in Dictator Games (Engel [2011]), our data allow us to report novel insights on social norm content and their strength in Dictator Games. Figure 1 shows that social appropriateness scores display a non-monotonic pattern: giving nothing is seen as socially inappropriate, whereas giving around half is rated as most appropriate. For actions involving transfers above half of the endowment, social appropriateness remains high—albeit declining somewhat relative to the 50-50 split. Despite such high appropriatenesses of giving more than half, very few subjects do so. This provides an early indication that norm appropriateness does not fully explain behavior. Social norm strength can explain why, as it is high for sending actions between 0 to 0.5 and then rapidly decreases for actions that would send more than 0.5.

4.2 Structural estimates

We now report the estimates of all utility models discussed in the previous sections. Table 2 provides an overview of all parameters estimated across all models. The first estimated model is the one considering selfish motives only (Eq. 1). From this benchmark model, we learn that dictators put a positive weight on their own payoffs and tend to choose the option that maximizes their monetary payoffs (average $\delta = 0.12$, 95%CI=[0.11,0.13]). However, while providing a good benchmarking, this model specification excludes social preferences or social influence.

When including social preferences (as in Eq. 2), results show that individuals are generally inequality averse. From estimates reported in Table 2, dictators typically place positive weight on others' payoffs when earning more (average $\rho = 0.50$, 95%CI=[0.49,0.51]) but they give a negative weight to other's payoffs when earning less (average $\sigma = -0.09$, 95%CI=[-0.13, -0.05]). These

⁴We do not provide a unique estimation from pooling all papers together given the heterogeneity in action spaces across experimental settings (e.g., ten possible actions vs. twenty possible actions), which may affect the reliability of estimates.

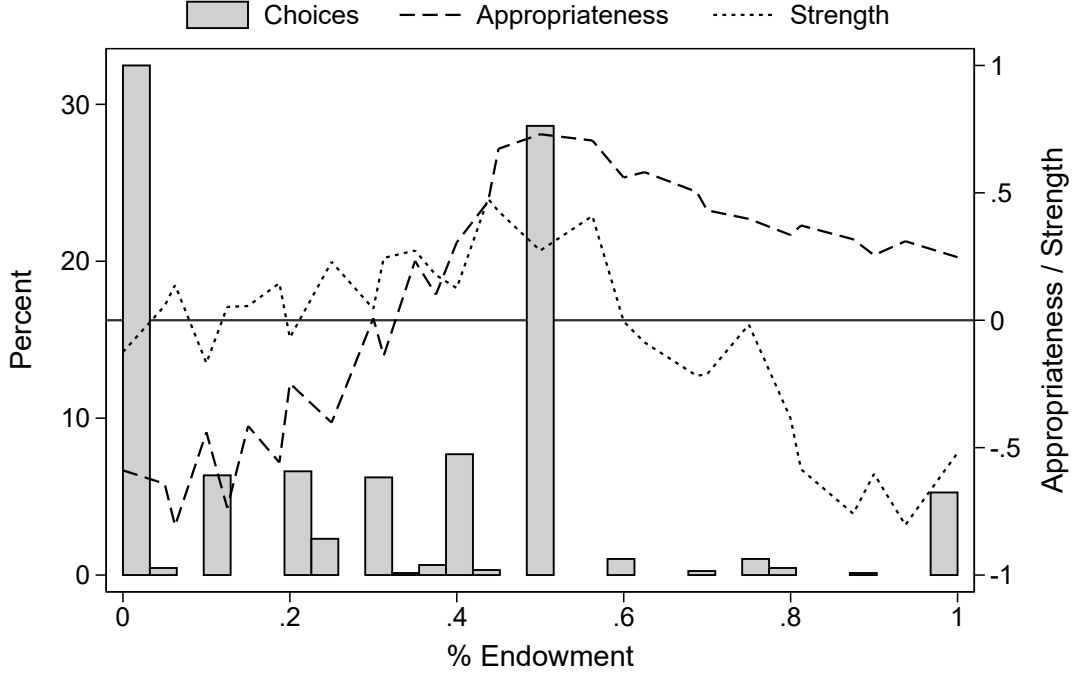
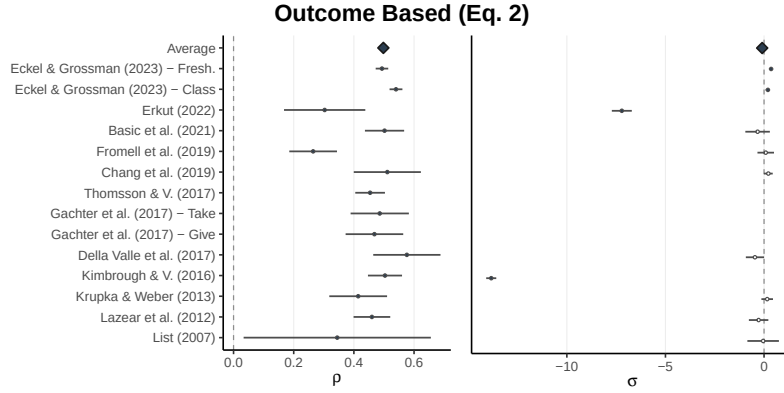


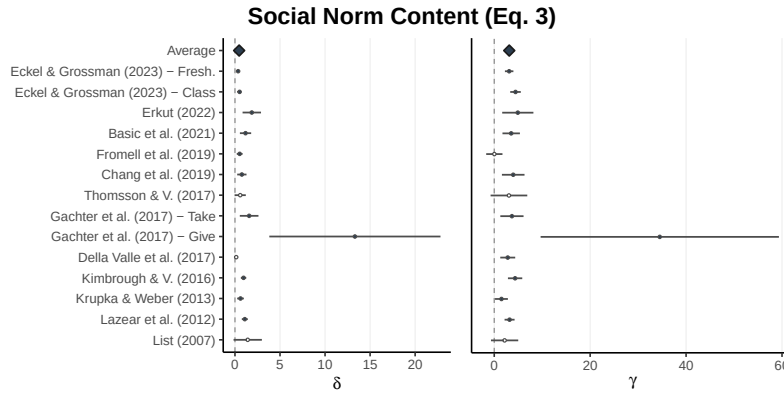
Figure 1: Frequency distribution of amounts given in the DG as percentage of initial endowment. Mean appropriateness is the average of appropriateness score in the social norm elicitation task (Krupka and Weber [2013]), while Norm strength is the (inverted) standard deviation of appropriateness scores for each action, normalized between -1 and 1, with higher values indicating a more strongly shared (less dispersed) norm.

results indicate that the representative Dictator cares about the Receiver’s payoff but only when they are ahead and not when they are behind. However, the results concerning σ are sensitive to the inclusion of two experiments (Erkut [2022], Kimbrough and Vostroknutov [2016]) in which no dictator gave more than half of their endowment, providing as a consequence very negative estimates of σ . With these papers removed, the estimate of σ becomes positive (average $\sigma = 0.31$, 95%CI=[0.26, 0.35]), suggesting that individuals are generally altruistic by giving a positive weight on other’s payoffs, although a lower one when in a disadvantageous position (i.e., giving more than half of the endowment). The sign of this outlier-excluded estimate ($\sigma = 0.31$, 95%CI=[0.26, 0.35]) is directionally consistent with Carpenter and Robbett [2022], who report a small positive behind-weight ($\hat{\alpha} = 0.076$) under the same utility specification, though the magnitudes differ.

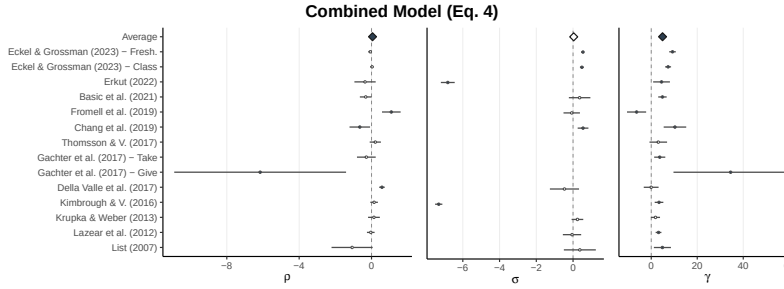
Turning to the models including social norm content (Eq. 3), estimates suggest that individuals are strongly concerned about social norms (average $\gamma = 3.13$, 95%CI=[2.71, 3.55]). These results hold unchanged even when estimating Eq. 4, encompassing both social norms and social preferences (average $\gamma = 4.92$, 95%CI=[4.35, 5.48]). Interestingly, by comparing social preferences model (Eq. 2) with the model specification also accounting for social norms (Eq. 3), the magnitude of social preferences weights shrinks, indicating that most of the behavior previously attributed to inequity



(a) Outcome Based (Eq. 2)



(b) Social Norm Content (Eq. 3)



(c) Combined Model (Eq. 4)

Figure 2: Fixed-effects (IVW) pooled estimates for Equations 2–4. Diamond: IVW average with 95% CI. Circles: individual-study estimates; filled = $p < 0.05$, hollow = n.s.

aversion in the model without norms is actually due to individuals' willingness to follow norms (average $\rho = 0.06$, 95%CI=[0.00, 0.11], average $\sigma = -0.04$, 95%CI=[-0.08, 0.00]). This pattern is consistent with Nunnari and Pozzi [2026], whose meta-analytic estimate of advantageous-inequality

aversion ($\hat{\beta} = 0.326$) lies below our norm-free estimate ($\rho = 0.50$) but above our norm-adjusted one ($\rho = 0.06$), suggesting that studies which do not control for norm compliance tend to attribute norm-driven behaviour to distributional preferences. Carpenter and Robbett [2022] document the same pattern within a single study, finding that the equivalent preference weight declines from $\hat{\beta} = 0.336$ to 0.263 once social appropriateness enters the utility function — a qualitatively identical collapse to the one we observe at the meta-analytic level. Table A3 in the appendix provides a side-by-side comparison of our estimates with those of Carpenter and Robbett [2022] and Nunnari and Pozzi [2026].

Social norm strength. We now turn to the estimates from the full model including both norm content and norm strength (Eq. 5). Model estimates corroborate results from previous models. Individuals generally care about adhering to the most socially appropriate actions (average $\gamma = 5.70$, 95%CI=[4.94, 6.45]). Holding constant the appropriateness of a given action, they also place positive weight on norm strength itself, i.e. they prefer choices for which there is greater consensus on their appropriateness level (average $\eta = 2.62$, 95%CI=[1.40, 3.85]). The role of social norm strength can be even more clearly seen when it is allowed to interact with content (Eq. 6): stronger social norms amplify norm-following behaviour, binding individuals more closely to the socially prescribed action, while weaker norms function more like non-binding suggestions (average $\nu = 3.90$, 95%CI=[1.75, 6.05]). Overall, results from both models demonstrate the important effects of social norm strength in translating what people know should happen into action. Put differently, when studying the effect of norms onto individuals’ behaviour, we should not only care about the *modal* answer, but also about the *variability* surrounding it.

Results from all models included in our conceptual framework are robust to the use of random-effects (DerSimonian-Laird) pooling in place of Inverse Variance Weighted (IVW) fixed-effects; see Appendix A3 for the random-effects estimates. Furthermore, design heterogeneity across studies — in particular, differences in whether social norm beliefs were elicited from the same or a different sample of participants — does not systematically affect norm content or norm strength measures; see Appendix A4 for details.

Marginal rates of substitution. A fundamental economic question is how individuals trade own’s material payoffs with those of others and with social appropriateness. Given our parametric estimates, we calculate the marginal rates of substitutions between one’s own and other’s payoffs, as well as the appropriateness of an action and the monetary payoff one has to give up. Overall, not considering the influence of social norms (i.e., Eq. 2), individuals are willing to sacrifice 0.99 tokens (i.e., $\rho/(1 - \rho)$) to increase the payoff of someone worse off by 1 token. This rate reduces to 0.06 tokens when accounting for social norms (i.e., Eq. 4). On the other hand, based on the estimates from social norms model (i.e., Eq. 3), individuals are ready to sacrifice an amount equivalent to 6.76 tokens (i.e., γ/δ) in the dictator game to take an action that is deemed one unit more appropriate.

4.3 Goodness of fit comparison

To assess the goodness of fit across model specifications, we compare the average estimates of AIC across papers for all model specifications (see Table A1, in the Appendix, for a treatment-wise comparison). A lower value of AIC indicates a better fit. Table 2 reports the average Akaike Information Criterion (AIC) for all models of utility. Model fit improves monotonically with the inclusion of richer behavioral components. The baseline Selfish model yields the highest AIC (466.21), indicating the poorest fit. Introducing social preferences significantly improves model performance

(AIC = 439.34), while models incorporating social norms perform even better. The Social Norm Content model achieves a lower AIC (411.86), and further gains are observed when jointly modelling social preferences and norm content (Combined Model, AIC = 397.61). Incorporating norm strength yields additional improvements (Social Norm Strength model, AIC = 386.38), and the Full model, which also allows norm content and norm strength to interact, achieves the best fit (AIC = 386.09). This progressive decrease in AIC underscores the incremental explanatory power of social preferences, social norms, and norm strength, with the Full model capturing the most nuanced individual motivations behind behavior in the Dictator Game.

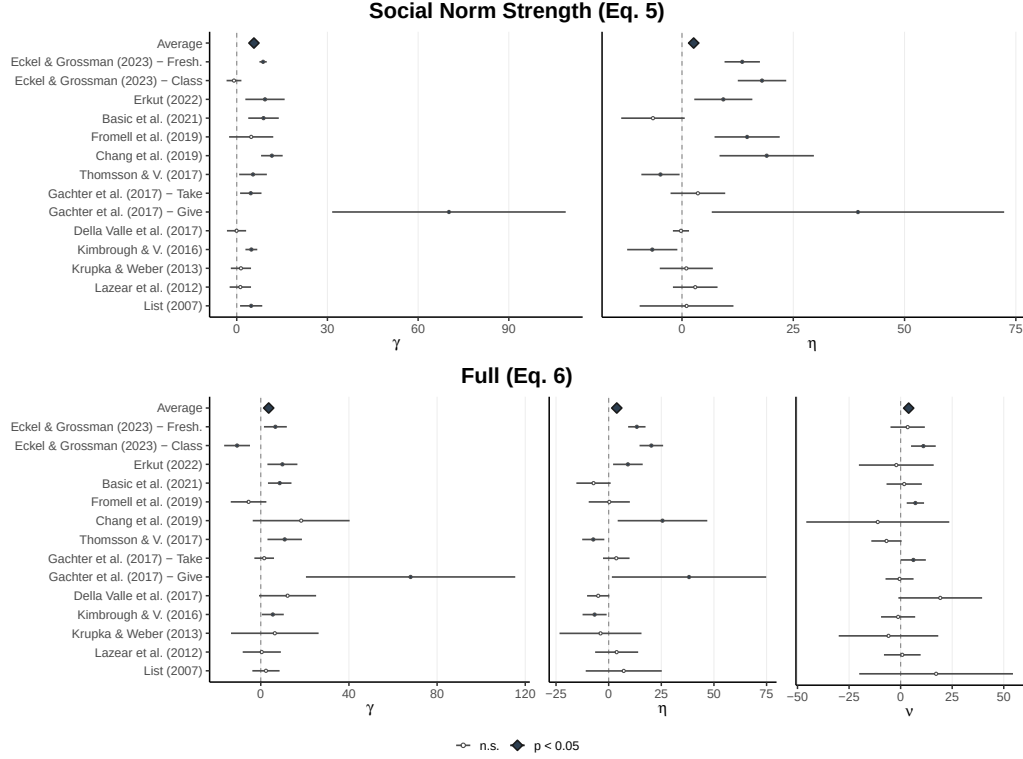


Figure 3: Fixed-effects (IVW) pooled estimates for the Norm Strength models (Equations 5–6). Diamond: IVW average with 95% CI. Circles: individual-study estimates; filled = $p < 0.05$, hollow = n.s.

5 Discussion

Our findings have important implications for both theory and policy. First, they suggest that models of prosocial behaviour should explicitly account for variation in norm strength. Norms influence behaviour most effectively when they are sufficiently shared and unambiguous, and ignoring this dimension risks overstating their explanatory power. A direct implication concerns situations in which choices are widely considered inappropriate, such as low sharing in the Dictator Game. In these contexts, weak norms expand moral wiggle room by relaxing the perceived normative constraint, making self-interested deviations easier to justify. This mechanism parallels the literature on moral

Model	δ	Social preferences		Social Norms			AIC
		ρ	σ	γ	η	ν	
Selfish (Eq. 1)	0.12 [0.11, 0.13]						466.21
Outcome Based (Eq. 2)		0.50 [0.49, 0.51]	-0.09 [-0.13, -0.05]				439.34
Social Norm Content (Eq. 3)	0.46 [0.40, 0.53]			3.13 [2.71, 3.55]			411.86
Combined Model (Eq. 4)		0.06 [0.00, 0.11]	0.04 [-0.01, 0.08]	4.92 [4.35, 5.48]			397.61
Social Norm Strength (Eq. 5)		0.02 [-0.04, 0.07]	-0.28 [-0.37, -0.18]	5.70 [4.94, 6.45]	2.62 [1.40, 3.85]		386.38
Full (Eq. 6)		0.20 [0.07, 0.34]	0.28 [0.17, 0.38]	3.59 [1.75, 5.43]	3.85 [1.97, 5.73]	3.90 [1.75, 6.05]	386.09

Table 2: Estimated parameters across utility specifications.

licensing and moral opportunism, which shows that individuals exploit ambiguity in moral standards to legitimize norm violations (Dana et al. [2007], Mazar et al. [2008], van Baar et al. [2019]). Our findings are also consistent with theoretical models in which moral opportunism arises endogenously from norm-dependent decision making under uncertainty (Merguei et al. [2022]). From an applied perspective, this suggests that framing interventions that reduce ambiguity and clarify social agreement may be particularly effective in increasing prosocial behaviour.

A second implication concerns the interpretation of low norm compliance. Our results show that observing weak compliance with a norm does not uniquely identify the underlying mechanism. Low adherence may reflect the presence of a norm that is endorsed in terms of content but weak in terms of strength (e.g., keeping all), or it may reflect a situation in which no shared norm exists in the first place (e.g., giving all). These cases require different policy responses. When a norm exists but is weak, interventions that strengthen coordination on expectations—such as providing information about others’ behavior or emphasizing social approval—are likely to be effective. When norms are absent, more structural or educational interventions may be required. Measuring both norm content and norm strength therefore enables more precise diagnoses of noncompliance and more targeted intervention strategies.

A third implication relates to the debate between outcome-based and norm-based explanations of prosocial behavior. Our results indicate that a substantial share of behavior traditionally attributed to social preferences can be more parsimoniously explained by normative motivations. From a policy perspective, this suggests that updating beliefs about others’ behavior and social approval may generate larger and faster behavioral changes than interventions aimed at reshaping deep preferences, especially when norms are perceived as strong.

Finally, our study contributes to a growing literature on norm strength from a micro-level perspective (Szekely et al. [2025, 2021]). Previous research has largely approached norm strength from a macro or cultural standpoint, for instance through concepts such as tightness versus looseness of societies (Gelfand et al. [2011]). In line with previous work, our meta-analytic approach shows that norm strength also operates at the individual level, and that variability in perceived consensus systematically predicts behavior. By explicitly separating norm content from norm strength,

our framework helps explain both individual differences in norm following and contextual fluctuations in compliance, providing a more comprehensive account of how social norms shape prosocial behaviour.

6 Conclusions

This paper provides meta-analytic evidence that behaviour in Dictator Games is jointly shaped by outcome-based social preferences and social norms, and that the influence of norms crucially depends on their strength. Across studies using incentivized elicitation of social appropriateness, models incorporating norm compliance consistently outperform purely outcome-based models, indicating that a significant portion of prosocial behavior is better explained by normative motivations than by distributional preferences alone.

Beyond norm content, our results highlight the central role of norm strength. Holding appropriateness constant, lower norm strength—our inverse measure of dispersion in appropriateness judgments—is systematically associated with lower compliance. Norms that are widely shared and clearly defined, such as near-equal splits, generate higher and more stable levels of prosocial behaviour. In contrast, ambiguous norms, such as giving nothing or giving everything, are characterized by greater behavioural variability and more frequent self-interested choices. These findings are consistent with a growing body of evidence showing that higher disagreement in normative beliefs weakens the behavioural impact of social norms (Szekely et al. [2021, 2025], Bicchieri and Demo [2024], Dimant [2023]). For instance, d’Adda et al. [2020] show that higher variance in normative beliefs leads to more heterogeneous contributions in Dictator games, while Dimant et al. [2024] demonstrate that perceived norm certainty increases compliance - the more certain a norm is (indicated by lower behavioral variance), the more likely individuals are to comply with it. Taken together, our results further corroborate the idea that compliance is not simply a function of what is considered appropriate, but also of how clearly and consistently that belief is shared.

Overall, our meta-analysis provides the initial systematic empirical support for the idea that individuals respond to differences in the strength of the social norms they face. By explicitly separating norm content from norm strength, we show that incorporating both dimensions is necessary to accurately capture the role of social norms in shaping prosocial behaviour.

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Appendices

A1 Additional Tables and Figures

Table A1: AIC comparison between models.

Paper	Treatment	Selfish (Eq. 1)	Outcome Based (Eq. 2)	Social Norm (Eq. 3)	Combined (Eq. 4)	Norm Strength (Eq. 5)	Full (Eq. 6)
List [2007]	Baseline	103.21	103.92	99.03	97.49	99.44	99.66
Lazear et al. [2012]	No Sorting	551.40	515.59	487.92	489.92	490.44	492.38
Krupka and Weber [2013]	Baseline	215.31	213.45	210.02	211.83	213.75	215.47
Kimbrough and Vostroknutov [2016]	DG	329.69	294.01	289.31	281.56	274.14	276.06
Della Valle and Ploner [2017]	Baseline	103.43	86.41	91.57	88.41	90.34	89.76
Gächter et al. [2017]	NoPeer Give	229.97	229.97	220.04	220.04	214.69	216.66
Gächter et al. [2017]	NoPeer Take	230.43	230.43	223.74	223.74	224.39	222.05
Thomsson and Vostroknutov [2017]	Baseline	251.79	251.79	250.77	250.77	247.73	247.55
Chang et al. [2019]	Neutrally-framed	255.01	248.21	221.74	203.70	186.07	187.58
Fromell et al. [2019]	Experiment 2 (Control)	416.89	417.33	418.89	410.61	388.90	381.34
Bašić and Verrina [2021]	Private - DG	536.68	488.77	457.10	454.57	451.19	452.91
Erkut [2022]	GENExperiment-Money	127.55	125.10	119.23	120.33	113.13	115.07
Eckel et al. [2023]	Class	1604.37	1416.39	1257.00	1216.35	1171.35	1163.59
Eckel et al. [2023]	Freshmen	1571.17	1529.45	1419.67	1297.19	1243.82	1245.20
Average		466.21	439.34	411.86	397.61	386.38	386.09

Note: AIC = Akaike Information Criterion; lower values indicate a better model fit. For each treatment (row), the lowest (best-fitting) AIC across the six specifications is shaded gray. Model specifications follow the equations in Table 2: Selfish (Eq. 1), Outcome Based (Eq. 2), Social Norm (Eq. 3), Combined (Eq. 4), Norm Strength (Eq. 5), Full (Eq. 6).

A2 Notes on papers

Despite our search criteria ensured comparability across studies, we had to make some ad-hoc choices when treating and analyzing data. This section documents treatment-specific features of the included studies that are relevant for interpreting the structural estimates reported in the main text.

- The appropriateness ratings used to construct the norm-based measures for List [2007] and Lazear et al. [2012] were not collected by the original authors; we use the elicitation conducted by Krupka and Weber [2013] for the same action space.
- In Della Valle and Ploner [2017] (Baseline), the dictator’s endowment was determined by a real-effort task, so appropriateness ratings were elicited for a fixed set of seven actions (0–6 tokens) that does not necessarily coincide with the actual action space available to each participant (which ranged from 5 to 15 tokens). We map appropriateness ratings onto each participant’s actual action space using the percentage of the endowment transferred.
- Fromell et al. [2019] (Experiment 2, Control) implemented a modified Dictator Game with multiple dictators and a single receiver. The control condition we use nonetheless closely resembles a standard Dictator Game, since one dictator’s decision is selected at random for payment and dictators receive no information about, nor are influenced by, the other dictators.
- The baseline Dictator Game in Gächter et al. [2017] (NoPeer Give and NoPeer Take) and Thomsson and Vostroknutov [2017] (Baseline) had a truncated action space: dictators received an endowment of 20 tokens but could transfer at most 10. Because it is infeasible by design for a dictator to end up in a disadvantageous position, we estimate only the advantageous-inequality-aversion parameter ρ and not σ for these treatments, using the full 20-token endowment.
- Gächter et al. [2017] additionally varied the dictator’s initial endowment using the strategy method (Selten, 1967), allowing dictators to take from, rather than give to, the receiver (NoPeer Take). We restrict attention to the subset of decisions in which the dictator held the full endowment, which corresponds to a standard Dictator Game.
- In Kimbrough and Vostroknutov [2016] (DG) and Erkut [2022] (GENExperiment-Money), no dictator transferred more than half of the endowment. As a result, σ is estimated from very few observations and its estimate is highly negative; this pattern is common across Dictator Game studies more generally.
- Coefficient estimates for γ in Eckel et al. [2023] differ from those reported in the original paper because of differences in sample construction: we restrict attention to each participant’s first-played Dictator Game, whereas the original study pools first- and second-played decisions.

A3 Random-Effects Estimates

The coefficient plots in the main text show inverse-variance weighted (IVW) fixed-effects averages, which assume a single true parameter value shared across all studies. As a robustness check, we re-estimated each pooled coefficient using the DerSimonian–Laird random-effects (RE) estimator, which allows for genuine between-study heterogeneity. The RE estimator adds an estimated between-study variance $\hat{\tau}^2$ to the sampling variance of each study before computing the weighted average, so studies with high heterogeneity receive less weight and the resulting confidence interval is wider.

Figures A1 and A2 replicate the main coefficient plots under the RE estimator. For most parameters the point estimates shift only modestly relative to the FE averages. The largest divergence occurs for σ (disadvantageous inequality aversion), which is a design artifact: in the majority of treatments no dictator transfers more than half the endowment, so the coefficient is estimated from very few observations and is numerically unstable across studies. For the remaining parameters, the RE and FE estimates are directionally consistent.

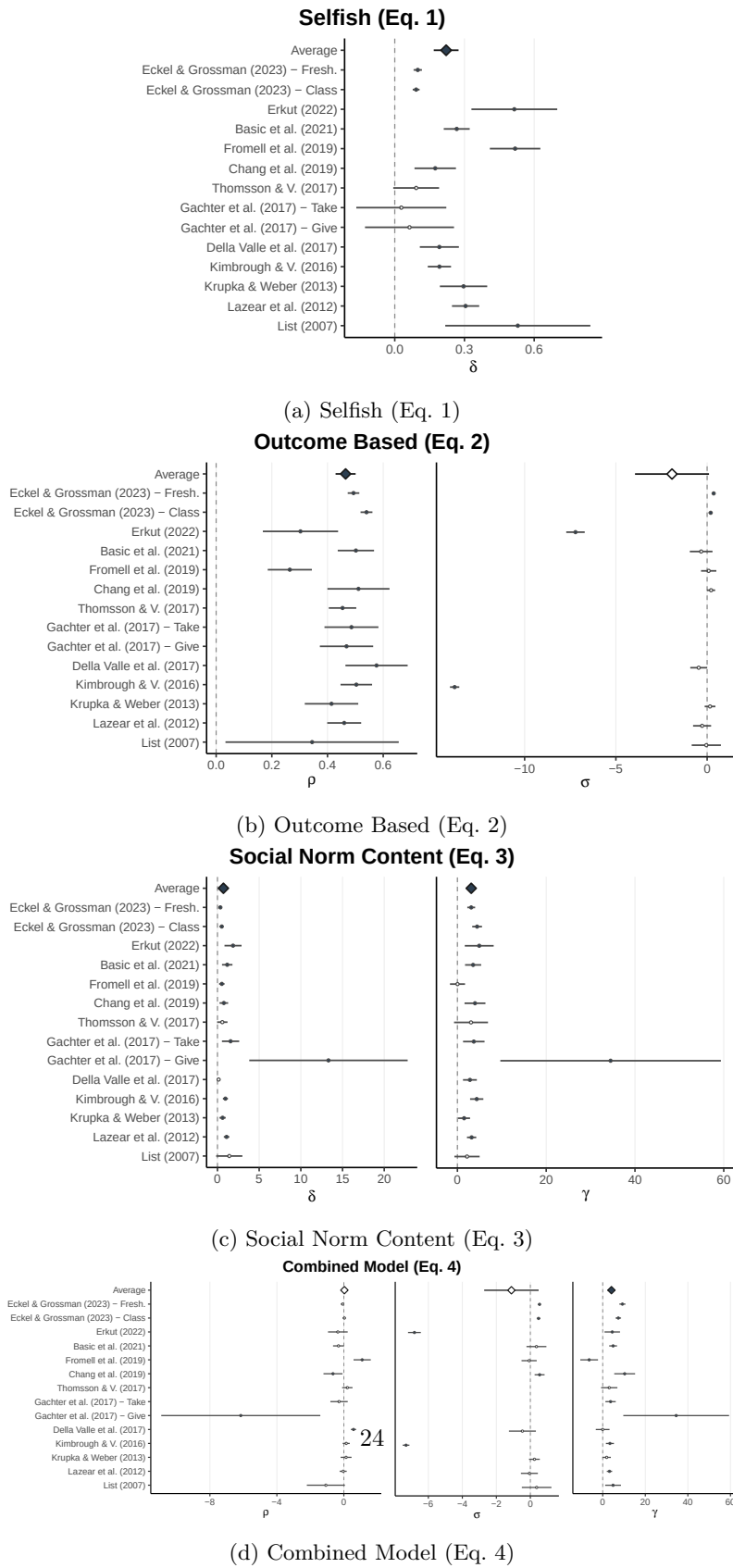


Figure A1: Random-effects pooled estimates for Equations 1–4. Diamond: RE average with 95% CI. Circles: individual-study estimates; filled = $p < 0.05$, hollow = n.s.

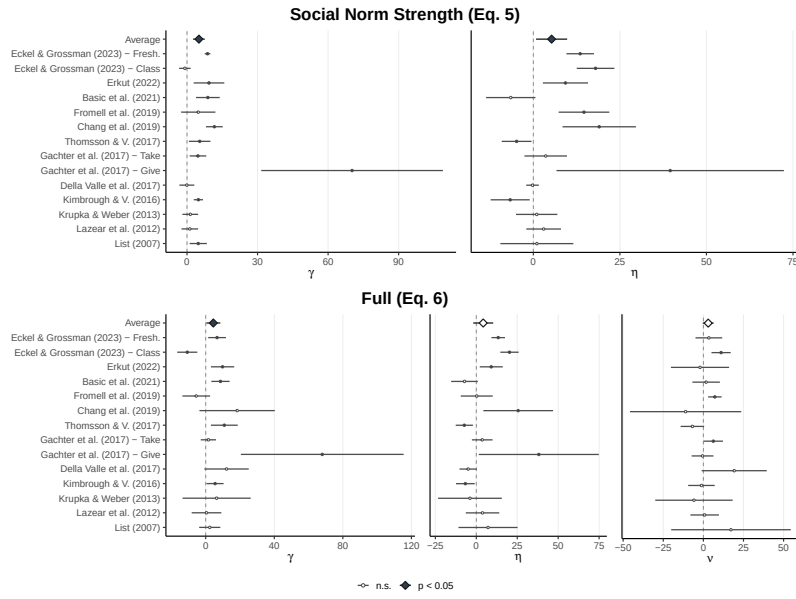


Figure A2: Random-effects pooled estimates for the Norm Strength models (Equations 5–6). Diamond: RE average with 95% CI. Circles: individual-study estimates; filled = $p < 0.05$, hollow = n.s.

A4 Design Heterogeneity

The 14 treatments included in the structural estimation differ in whether the appropriateness ratings used to construct norm content (N_k) and norm strength (S_k) were elicited from the same individuals who made the dictator-game decisions (*Within*, $k = 5$ treatments) or from an independent sample of raters (*Between*, $k = 9$ treatments). This distinction matters in principle: ratings elicited from the same individuals who later choose could be subject to consistency motives or anchoring across actions, while ratings from an independent sample avoid that channel but introduce additional between-sample sampling noise. Given that our sample includes only 14 treatments, formally testing for a difference between the two designs would have very little power, so we report two descriptive checks instead of a formal test.

First, Figures A3 and A4 show the within-treatment distribution of mean appropriateness and of norm strength S_k across all rated actions, split by belief-sample design. Neither measure shows a systematic shift between Within- and Between-sample treatments; both designs span a similar range and neither one is consistently higher or lower.

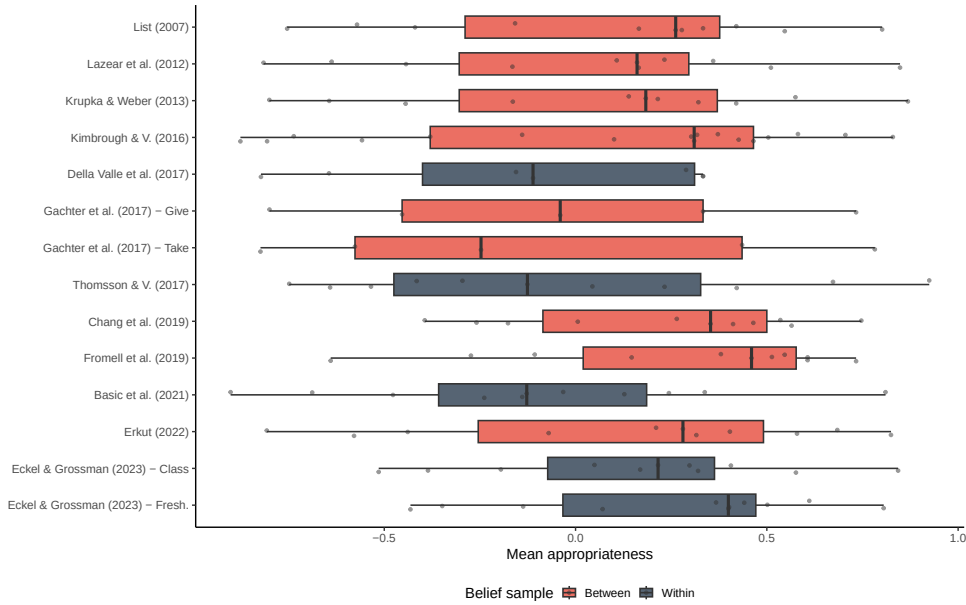


Figure A3: Distribution of mean appropriateness ratings across treatments, by belief-sample design. Each point is one rated action; boxplots summarize the within-treatment distribution.

Second, Table A2 re-estimates each norm-dependent model (Eq. 3–6) separately on the Within and Between subsamples, pooling coefficients within each subsample via the same inverse-variance-weighted procedure used for the main estimates. The sign and rough magnitude of γ , η , and ν are preserved across both subsamples, suggesting the structural role of appropriateness and norm strength is not an artifact of how the beliefs were elicited; differences in absolute AIC reflect the difference in subsample size rather than fit quality, since AIC is not comparable across samples of different size.

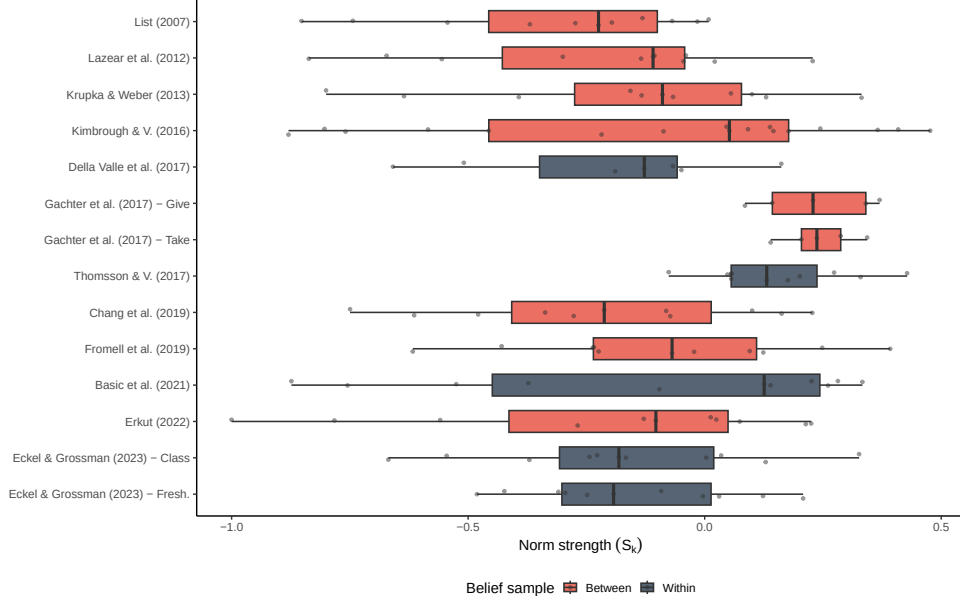


Figure A4: Distribution of norm strength S_k across treatments, by belief-sample design. Each point is one rated action; boxplots summarize the within-treatment distribution.

Model	Belief sample	Social preferences			Social Norms			AIC
		δ	ρ	σ	γ	η	ν	
Social Norm Content (Eq. 3)	Within ($k = 5$)	0.37 [0.29, 0.44]			3.49 [2.90, 4.08]			695.22
Social Norm Content (Eq. 3)	Between ($k = 9$)	0.86 [0.70, 1.01]			2.77 [2.17, 3.36]			254.43
Combined Model (Eq. 4)	Within ($k = 5$)		0.06 [0.00, 0.12]	0.53 [0.48, 0.57]	6.82 [6.02, 7.62]			661.46
Combined Model (Eq. 4)	Between ($k = 9$)		0.04 [-0.08, 0.16]	-3.50 [-3.62, -3.37]	3.06 [2.27, 3.86]			251.02
Social Norm Strength (Eq. 5)	Within ($k = 5$)				6.21 [5.24, 7.19]	2.11 [0.66, 3.56]		640.89
Social Norm Strength (Eq. 5)	Between ($k = 9$)				4.93 [3.74, 6.12]	3.94 [1.63, 6.25]		244.99
Full (Eq. 6)	Within ($k = 5$)				3.92 [1.08, 6.76]	4.96 [2.58, 7.33]	3.90 [0.29, 7.50]	639.80
Full (Eq. 6)	Between ($k = 9$)				3.36 [0.95, 5.77]	1.98 [-1.11, 5.07]	3.90 [1.22, 6.58]	245.14

Table A2: Pooled (IVW) coefficient estimates split by whether the appropriateness ratings used to construct N_k/S_k came from the same sample as the dictator-game choices (Within, $k = 5$ treatments) or an independent sample (Between, $k = 9$ treatments). Only models involving N_k/S_k are shown; Eq. 1–2 do not depend on these measures.

A5 Comparison with Existing Structural Estimates

Table A3 places our representative-agent estimates alongside those of Carpenter and Robbett [2022] and Nunnari and Pozzi [2026], the two closest benchmarks in the literature. The ahead-weight (ρ

$/ \hat{\beta}$) is directly comparable across all three papers; the behind-weight ($\sigma / \hat{\alpha}$) requires care because Nunnari and Pozzi [2026] use the Fehr-Schmidt parameterization, in which $\hat{\alpha} > 0$ denotes disutility from being behind — a positive $\hat{\alpha}$ in that framework corresponds to a negative σ in the Charness-Rabin framework used here and by Carpenter and Robbett [2022]. Norm sensitivity (γ) is not estimated by Nunnari and Pozzi [2026].

Table A3: Representative-agent estimates across studies

Parameter	This paper		Carpenter & Robbett (2022)		Nunnari & Pozzi (2026)
	No norms	With norms	No norms	With norms	(no norms)
Ahead weight ($\rho / \hat{\beta}$)	0.50 [0.49, 0.51]	0.06 [0.00, 0.11]	0.336***	0.263***	0.326 [0.254, 0.398]
Behind weight ($\sigma / \hat{\alpha}$)	−0.09 [−0.13, −0.05]	0.04 [−0.01, 0.08]	0.076***	0.078***	0.533 [†] [0.311, 0.755]
Norm sensitivity (γ)	—	4.92 [4.35, 5.48]	—	26.53***	—

Notes: 95% confidence intervals in brackets. For Carpenter and Robbett [2022], confidence intervals are not reported in the original table; *** denotes $p < 0.01$. Our estimates use inverse-variance-weighted fixed-effects pooling; Nunnari and Pozzi [2026] use a three-level random-effects model.

[†] Nunnari and Pozzi [2026] estimate the Fehr-Schmidt model, in which $\hat{\alpha}$ captures disadvantageous inequality aversion (disutility from being behind). A positive $\hat{\alpha}$ in that framework corresponds to a negative σ in the Charness-Rabin framework used by this paper and Carpenter and Robbett [2022]; the two estimates are therefore directionally consistent.